

How to use AETS E-BOX



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MANUAL MODE

How to navigate in the menu using encoder:

Navigation of the menu is done by using encoder. Encoder has an advantage in the fact that it can rotate freely to either direction without stopping. Encoder that is used is equipped with a switch, that can be used by pressing the encoder. This is used to select items.

Where you are in the menu is indicated by the cursor in the first collum from the left.

If you wish to go into the previous menu, select the option < Return on top of each menu.

Cursor:



Go down



Go up



Select



Return:



INIT:

This screen appears on power-up of the device.

INIT	A u t o m a t e d																			
	E l e c t r i c a l																			
	T e s t i n g																			
	S y s t e m v 1 . 0																			

Calibration screen:

After power-up the device needs to self-calibrate its current zero offset. This screen tells you to remove all testing cables and DUT'S. After that you need to press encoder for the calibration. Successful calibration is indicated by buzzer and screen going to main menu.

Calibration 1	Z E R O C U R R E N T C A L .																			
	P L E A S E D I S C O N N E C T																			
	A L L C A B L E S A N D L O A D S																			
	P R E S S T O C A L I B R A T E																			
Calibration 2	C A L I B R A T I N G . . .																			

Main Menu:

This is the main screen of the menu. From here you can go to every sub-menu to set up tests, load profiles, set communication etc.

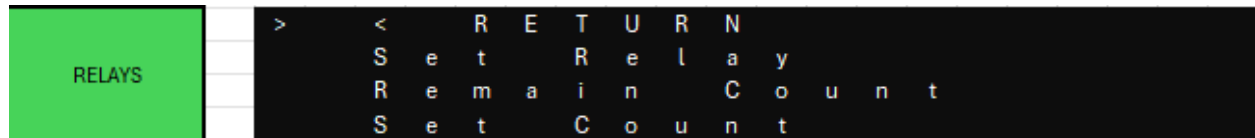
These are the main sub-menus:

- Start test
- Relays
- Mosfets
- Trigger
- Connectivity
- Profiles
- Settings
- Info

MENU TOP	> S t a r t T e s t																			
	R e l a y s																			
	M o s f e t s																			
	T r i g g e r																			
MENU BOT	> C o n n e c t i v i t y																			
	P r o f i l e s																			
	S e t t i n g s																			
	I n f o																			

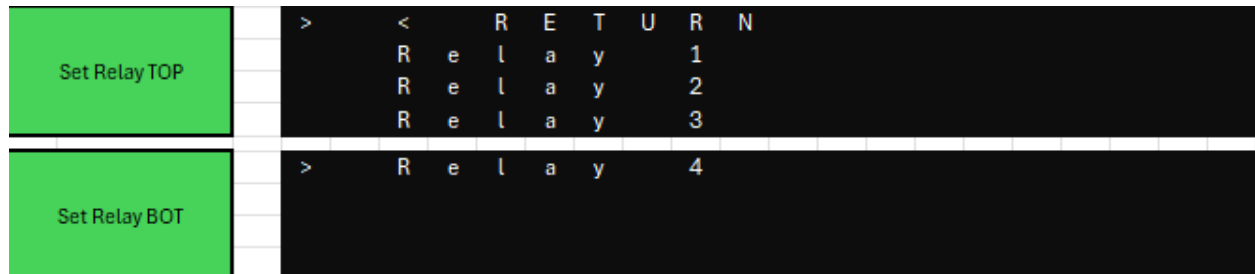
Relays:

Here you can set up to 4 relays.



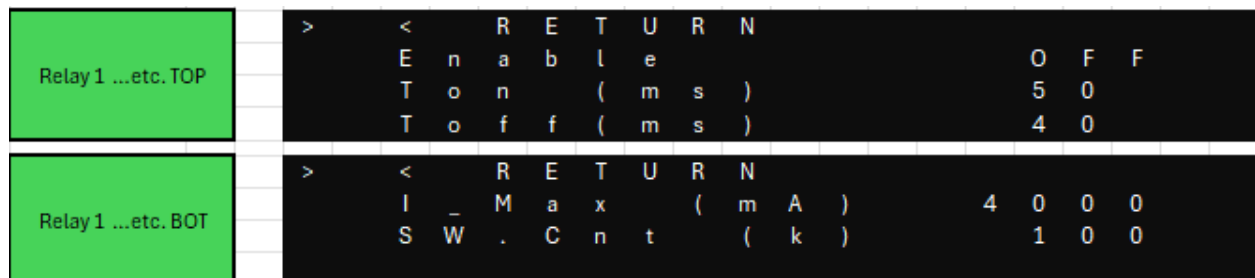
Set Relay:

First you choose which relay you want to configure.



Relay x:

This is the main configuration screen for relays.



- Enable [ON|OFF]
 - This will enable the relay, and it will be included in the switching sequence
- Ton (ms) [0 – 1000000]
 - Time the relay will be on
- Toff (ms) [0 – 1000000]
 - Time the relay will be off
- I_max (mA) [0 – 4000]
 - This value sets the maximum current that can pass through the relay. If current is above this value for 3 consecutive times, the test will go into fail.
- SW.Cnt (k) [0 – 1000000]
 - How many times will relay switch on and off. This value is in the thousands!!!

Remain Count:

This screen shows you how many switches each relay has remaining. This is read-only information

Remain Count TOP	>	<	R	E	T	U	R	N											
		R	e	l	a	y	1	(	k	)	1	0	0	0	0				
		R	e	l	a	y	2	(	k	)	2	5	4	4	3				
		R	e	l	a	y	3	(	k	)	5	0	0	0	0				
Remain Count BOT	>	R	e	l	a	y	4	(	k	)	1	0	0	0	0				

Set Count:

This screen is used to setup switches for new relays.

Set Count TOP	>	<	R	E	T	U	R	N											
		R	e	l	a	y	1	(	k	)	3	0	0	0					
		R	e	l	a	y	2	(	k	)	3	0	0	0					
		R	e	l	a	y	3	(	k	)	3	0	0	0					
Set Count BOT	>	R	e	l	a	y	4	(	k	)	3	0	0	0					

Trigger:

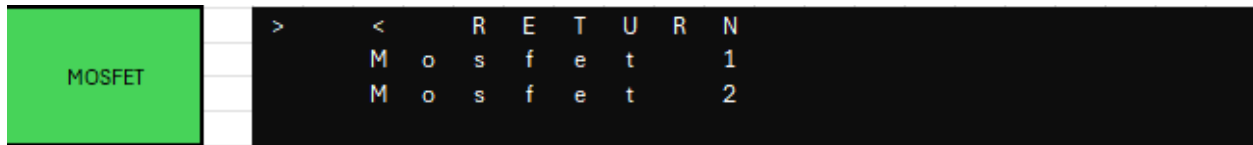
Trigger menu allows you to activate trigger output. You can choose with which relay it will activate. Trigger is activated when relay turns on

- Enable (ON|OFF) – Enables trigger in the testing
- Relay Ch. – Choose which relay channel will the trigger follow

TRIGGER	>	<	R	E	T	U	R	N											
		E	n	a	b	l	e				O	F	F						
		R	e	l	a	y	C	h	.				2						

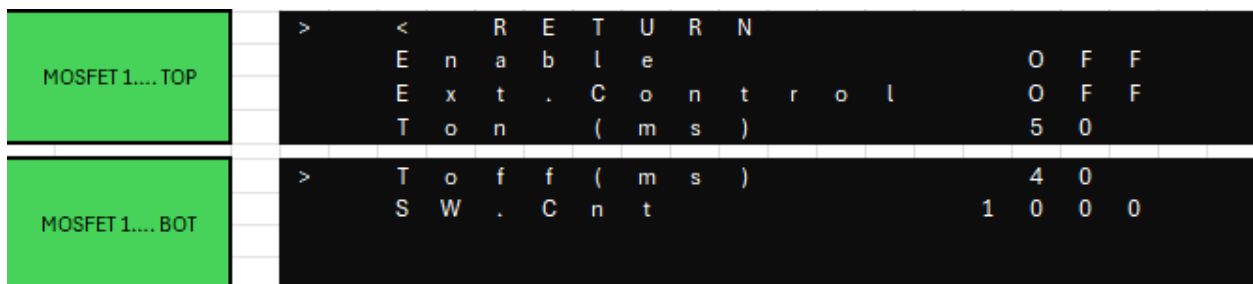
MOSFETs:

Here you can choose which channel you want to configure.



Mosfet x:

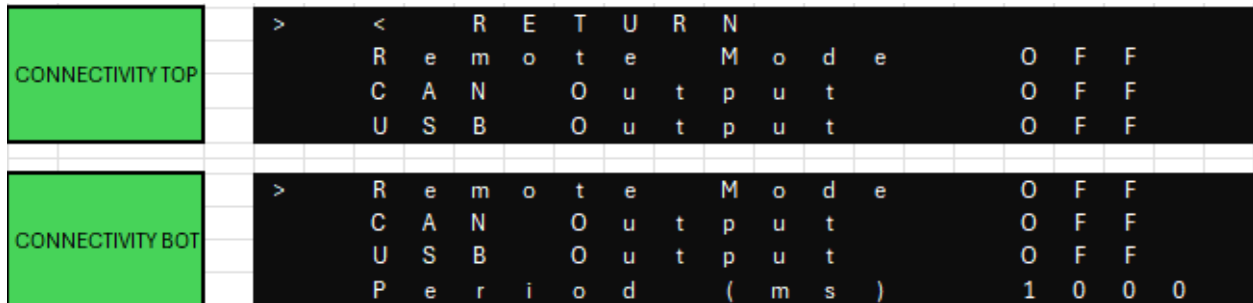
This is the main configuration screen for mosfets.



- Enable [ON|OFF]
 - This will enable the mosfet, and it will be included in the switching sequence
- Ext.Control [ON|OFF]
 - This changes driving of the MOSFET from MCU control to external control. External control is utilized by BNC connector. This allows you to connect waveform generator to the device and bypass MCU control. Be aware to connect BNC cable to the right channel you want to use.
- Ton (ms) [0 – 1000000]
 - Time the mosfet will be on
- Toff (ms) [0 – 1000000]
 - Time the mosfet will be off
- SW.Cnt [0 – 1000000]
 - How many times will mosfet switch on and off.

Connectivity:

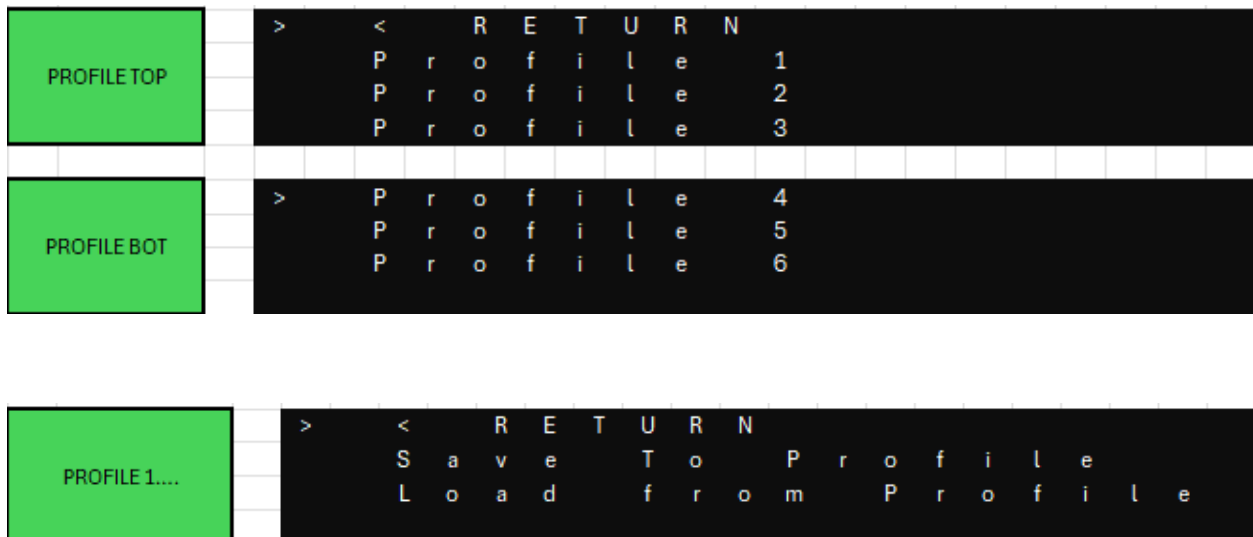
In this menu you can configure communication options.



- Remote Mode [ON|OFF]
 - This allows you to switch to remote mode, where you can control the device via commands using Virtual Serial Port.
- CAN output [ON|OFF]
 - If switched on it will output data in 2 frames during the test from DSUB-9 connector
- USB Output [ON|OFF]
 - If switched on it will output data into Serial Port during the test from USB-C connector
- Period (ms) [200 - 100000]
 - Determines period of sending output data during test.

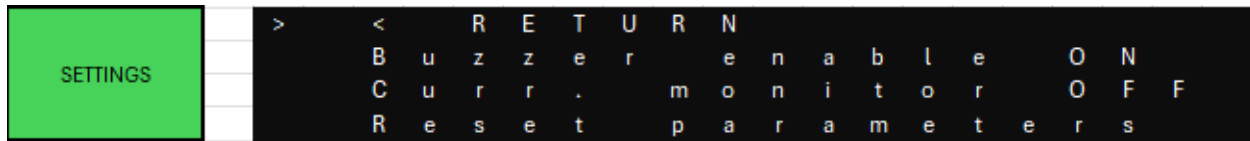
Profile:

Here you can save your test settings or load them up. The device has 6 profiles available.



Settings:

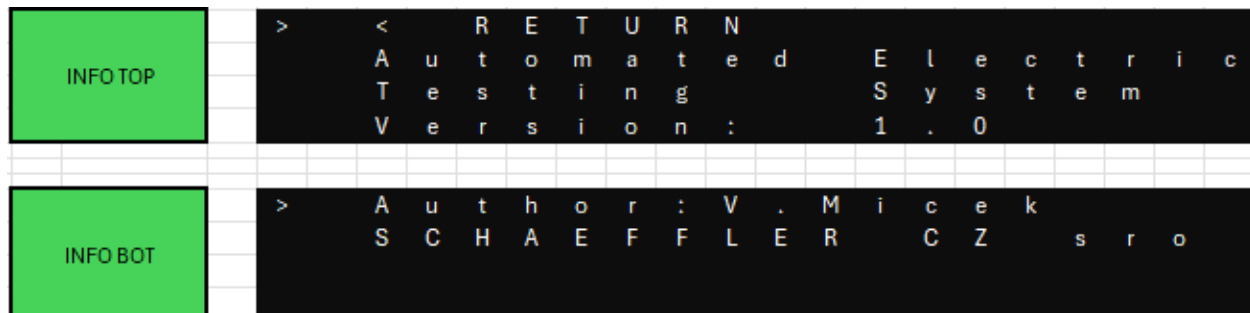
Here are basic system settings.



- Buzzer enable [ON|OFF]
 - Turn on/off sound feedback of the device.
- Curr. monitor [ON|OFF]
 - You can turn monitoring on/off for the test. This is used for internal control of relay -> overcurrent / zero current faults
- Reset parameters
 - This refreshes all system values to default state.

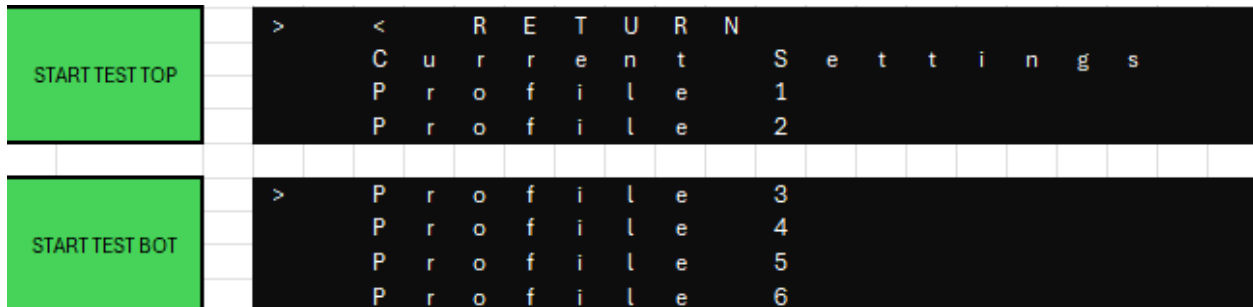
INFO:

This menu displays information about the device.



Start Test:

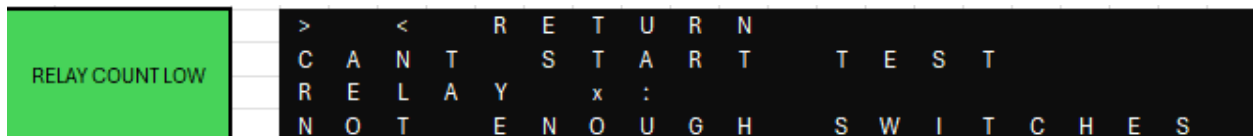
Here you can choose which configuration of test you want.



- Current Settings
 - Starts test with configured values.
- Profile 1-6
 - Starts test with configuration values loaded from the selected profile.

Relay Count Low:

If you want to start test where you have configured switches on relay that are larger than the remaining count of the relay, this screen will appear notifying you which relay needs to be reconfigured.



REMOTE MODE:

Connecting to the Device (USB-C)

To control the AETS E-BOX remotely or monitor its telemetry data, you must first establish a serial connection using the device's USB-C port.

Hardware Connection:

1. Plug a standard USB-C cable into the USB-C port on the AETS device.
2. Connect the other end to your host computer.
3. Your computer should automatically recognize the device and assign it a Virtual COM port (for example, COM3 on Windows or /dev/ttyUSB0 on Linux). You can verify the assigned port number in your computer's Device Manager or system settings.

Serial Port Configuration:

To ensure proper communication, open your preferred terminal emulator (such as PuTTY, Tera Term, or your automated testing software) and configure the serial connection with the following exact parameters:

- Baud Rate: 9600
- Data Bits: 8
- Stop Bits: 1
- Parity: None (0)

Initialization:

Once the physical connection is secure and the terminal is configured, you can verify the connection by sending a basic command like `gh` (to bring up the help menu). Remember that to execute full test controls, you must enable Remote Mode either via the physical device menu (Connectivity -> Remote Mode -> ON) or by sending the mode set remote command through your terminal.

The Remote Mode allows you to control the AETS device via commands using a Virtual Serial Port. To utilize the full command set, the device must actively be in this mode. You can enable it manually on the device display via the **Connectivity** menu, or by sending the allowed mode-switching command.

When the device is **not** in Remote Mode, only the mode command or the test stop command will be accepted. Any other commands sent will be rejected with an Error 18 (disabled) response.

Command Protocol & Responses

All commands sent to the AETS device will generate a standard acknowledgment (ACK) response. The standard structure for an expected response is:

ack <cmdName> <cmdId> <errorCode> <timestamp>

Error Codes Reference

If a command fails or is rejected, the <errorCode> will indicate the reason.

CODE	Description
0	Success / no error
1	Generic failure (e.g., low-level call failed)
2	Missing or invalid parameter (parsing error)
3	ACK error
4	NACK error
5	Initialization error
6	Busy / cannot accept command now (event queue full)
7	Timeout
8	Saturation / overflow condition
9	Parameter out of range
10	Command not found
11	NULL pointer or invalid pointer
12	Not implemented
13	Resource not found (e.g., load failure)
14	Not supported
15	Unsupported/invalid command or action
16	Overflow condition
17	CRC mismatch in command
18	Command disabled (e.g., blocked because device is not in REMOTE mode)

Command Directory

System & Information Commands

Command	Parameters	Expected Response / Action
gh	None	Multi-line help text, followed by ack gh <cmdId> 0 <timestamp>
gi	None	cmd gi <cmdId> <device info...> followed by ack gi <cmdId> 0 <timestamp>
mode get	None	cmd mode <cmdId> <state> followed by ACK
mode set	[manual remote test]	ACK with error 0 if the transition is allowed. A later evt state <state> event will follow.

Relay Commands

Command	Parameters	Expected Response / Action
relay get	None	cmd relay <cmdId> r1 r2 r3 r4 followed by ACK
relay set	[1-4] (Channel) [0 1] (State: 0=OFF, 1=ON)	ack relay <cmdId> 0 <timestamp>
rcnt get	[1-4] (Optional Channel)	Returns cycle counters for a single relay or all relays.
rcnt set	[1-4] (Channel) [value] (k-cycles)	Sets remaining cycles for a specific relay.
rcnt reset	None	Resets all relay counters to default values.
rcnt save	None	Saves current counter values to flash memory.
curr ma	[1-4] (Channel)	cmd current <cmdId> <ma> followed by ACK. Returns current measurement in mA.

MOSFET & MUX Commands

Command	Parameters	Expected Response / Action
mosfet get	None	cmd mosfet <cmdId> m1 m2 followed by ACK
mosfet set	[1 2] (Channel) [0 1] (State: 0=OFF, 1=ON)	ack mosfet <cmdId> 0 <timestamp>
mux get	None	cmd mux <cmdId> <mux1> <mux2> (0=INT, 1=EXT)
mux set	[1 2] (Channel) [0 1] (State: 0=INT, 1=EXT)	Selected mux channel switches (indicated by LED).

Test Control Commands

Command	Parameters	Expected Response / Action
test start current	None	ACK; device transitions to evt state TEST. Outputs follow current sequence.
test start profile	[1-6] (Profile Number)	ACK; device transitions to evt state TEST. Outputs follow selected profile.
test stop	None	ACK; device transitions to evt state REMOTE (or MANUAL if stopped manually).

Parameter & Profile Settings

Note: Using the param reset command resets all system parameters to their defaults.

Default states include Buzzer = ON (1), Connectivity Outputs = OFF (0), and Trigger = OFF (0) assigned to Channel 1.

Command	Parameters	Expected Response / Action
param relay get	[1-4] (Channel)	Returns parameters for the specified relay(s).
param relay set	[Channel 1-4] [Enable 0 1] [Ton ms] [Toff ms] [Imax mA] [Cycles k]	ACK
param mosfet get	[1 2] (Channel)	Returns parameters for the specified MOSFET.
param mosfet set	[Channel 1-2] [Enable 0 1] [ExtCtrl 0 1] [Ton ms] [Toff ms] [Cycles]	ACK. If ExtCtrl=1, MUX is forced to EXT and output is off.
param trigger get	None	Returns trigger enable state and assigned relay channel.
param trigger set	[Enable 0 1] [Channel 1-4]	ACK
param conn get	None	Returns states for Remote Mode, CAN output, USB output, and Period.
param conn set	[Remote 0 1] [CAN 0 1] [USB 0 1] [Period ms]	ACK; updates are reflected in the device menu.
param monitor get	None	Returns current monitoring state.
param monitor set	[Enable 0 1]	ACK; updates are reflected in the device menu.
param buzzer get	None	Returns buzzer enable state.
param buzzer set	[Enable 0 1]	ACK; updates are reflected in the device menu.
param reset	None	ACK; resets all current parameters to default.
param list profile	[1-6] (Profile Number)	ACK; lists every parameter stored in the selected profile.
save profile	[1-6] (Profile Number)	ACK; saves current configuration to specified flash profile.
load profile	[1-6] (Profile Number)	ACK; loads parameters from the specified flash profile.

Telemetry and Data Outputs

During testing, the device can broadcast telemetry data via USB or CAN bus, provided these outputs are enabled in the connectivity parameters.

USB Output Descriptions

When USB output is active, the device serial output will report test states and telemetry lines:

- **Running Telemetry:**
out <timestamp> r1 <remaining_k> <state> <current_ma> r2 <remaining_k> <state> <current_ma> r3 <remaining_k> <state> <current_ma> r4 <remaining_k> <state> <current_ma> m1 <state> m2 <state>
- **Test Blocked:**
Test blocked. Not enough switches left. Relay: x (Relay x lacks required cycles for the test).
- **Test Stopped:**
Test Stopped (Triggered manually or remotely).
- **Test Fail (Zero Current):**
Test Fail: Zero Current (Exceeded maximum steps with 0mA measured).
- **Test Fail (OverCurrent):**
Test Fail: OverCurrent (Exceeded maximum steps above maximum current threshold).
- **Test Done:**
Test Done (Test finished successfully).

CAN Bus Output Description

The CAN output broadcasts two distinct frames representing the state of the system.

Frame 1 (ID = 49)

Bits	Parameters	Description
0 .. 12	Current CH1	Relay 1 current in mA (13-bit clamped).
13 .. 24	Relay1 Count-k	Remaining switch count for Relay 1 in kilo-units.
25	Relay1 State	Relay 1 output state (0 = OFF, 1 = ON).
26	Relay1 Alive	Relay 1 health monitor (0 = remaining count $\leq$ 0, 1 = remaining count $>$ 0).
27 .. 39	Current CH2	Relay 2 current in mA (13-bit clamped).
40 .. 51	Relay2 Count-k	Remaining switch count for Relay 2 in kilo-units.
52	Relay2 State	Relay 2 output state (0 = OFF, 1 = ON).
53	Relay2 Alive	Relay 2 health monitor (0 = remaining count $\leq$ 0, 1 = remaining count $>$ 0).
54	MUX1 State	Multiplexer 1 state (0 = Internal, 1 = External).
55	MOSFET1 State	MOSFET 1 output state (0 = OFF, 1 = ON).
56 .. 63	Up-Counter	8-bit counter incremented every CAN frame.

Frame 2 (ID = 50)

Bits	Parameters	Description
0 .. 12	Current CH3	Relay 3 current in mA (13-bit clamped).
13 .. 24	Relay3 Count-k	Remaining switch count for Relay 3 in kilo-units.
25	Relay3 State	Relay 3 output state (0 = OFF, 1 = ON).
26	Relay3 Alive	Relay 3 health monitor (0 = remaining count $\leq$ 0, 1 = remaining count $>$ 0).
27 .. 39	Current CH4	Relay 4 current in mA (13-bit clamped).
40 .. 51	Relay4 Count-k	Remaining switch count for Relay 4 in kilo-units.
52	Relay4 State	Relay 4 output state (0 = OFF, 1 = ON).
53	Relay4 Alive	Relay 4 health monitor (0 = remaining count $\leq$ 0, 1 = remaining count $>$ 0).
54	MUX2 State	Multiplexer 2 state (0 = Internal, 1 = External).
55	MOSFET2 State	MOSFET 2 output state (0 = OFF, 1 = ON).
56 .. 63	Up-Counter	8-bit counter incremented every CAN frame.

TEST:

Screen during test. You can stop test manually by pressing the encoder or by stopping test remotely.

TEST TOP	> S t o p T e s t C o u n t L e f t : R 1 : 1 0 0 0 0 O N I : 3 7 0 0 R 1 : 0 5 0 0 0 O F F I : 8 2
TEST BOT	R 3 : * * * * * R 4 : 7 0 0 0 0 I : 5 5 1 M 1 : * * * * * M 2 : 0 6 0 0 0 O F F

- Enabled peripherals:
 - Relays:
 - Shows remaining count, state of the relay, current in mA
 - MOSFETs:
 - Show remaining count, state of the MOSFET
- Disabled peripherals:
 - If the periphery isn't used, it will fill the row with *****

Test end screens:

STOP TEST	> < R E T U R N T O M E N U T E S T S T O P P E D
TEST OK	> < R E T U R N T O M E N U T E S T F I N N I S H E D S U C C E S S F U L Y
ERROR MAX CURRENT TEST	> < R E T U R N T O M E N U T E S T E N D E D E R R O R : C U R R E N T E X C E D E D
ERROR 0 CURRENT TEST	> < R E T U R N T O M E N U T E S T E N D E D E R R O R : N O C U R R E N T F L O W